CLASS 6 — SCIENCE · HARD LEVEL FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–12) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. 100 g of a mixture contains 20 g of iron, 30 g of sand and 50 g of salt. After separation, the mass of sand obtained is expected to be: 1

(a) 20 g **(b)** 50 g **(c)** 30 g **(d)** 100 g

2. Out of a tree, a bird and a pebble, which one is non-living? 1

(a) pebble **(b)** tree
(c) bird **(d)** both bird and tree

3. Which of these actions improves soil fertility naturally? 1

(a) burning plastics **(b)** adding plastic **(c)** removing worms **(d)** adding compost

4. Mass of an object is 60 kg on Earth. On the Moon (gravity one-sixth), its mass is: 1

(a) 360 kg **(b)** 60 kg **(c)** 10 kg **(d)** 6 kg

5. Suppose two groups test the same hypothesis and get opposite results. What should be done first? 1

(a) toss a coin **(b)** believe one group
(c) ignore both **(d)** compare the methods and repeat carefully

6. In a forest, if all insects vanish, what is most likely to happen? 1

(a) many plants will fail to pollinate **(b)** animals will grow bigger
(c) forest will grow faster **(d)** all plants will grow faster

7. A paper rubbed with a food sample shows a translucent oily patch that stays even after drying. The food contains: 1

(a) starch **(b)** fat **(c)** fibre **(d)** sugar

8. To make a magnet from an iron bar by the single-touch method, the bar should be stroked: 1

(a) always in the same direction with the same pole of a magnet
(b) back and forth with both poles
(c) by hitting it with the magnet
(d) in a circle with the magnet

9. A student measuring a pencil places the 2 cm mark of the scale at one end and reads 12.5 cm at the other end. Length is: 1

(a) 2.0 cm **(b)** 12.5 cm **(c)** 14.5 cm **(d)** 10.5 cm

10. Why does a heavy ship made of steel float? 1

(a) steel is lighter than water **(b)** water is sticky
(c) ships have no weight **(d)** its shape displaces much water

11. The best conductor-insulator pair for a cooking pot and its handle is: 1

(a) aluminium and plastic **(b)** plastic and aluminium
(c) glass and iron **(d)** wood and plastic

12. A student claims that the water in a cup disappeared. The correct statement is: 1

(a) it vanished **(b)** it was destroyed
(c) it changed to vapour and mixed with air **(d)** it became oxygen

13. Why is alum added during water purification before filtration? 1

(a) it dissolves salt **(b)** it adds taste
(c) it kills all germs **(d)** it makes tiny particles clump and settle

14. A dry seed lies unchanged in a packet for months but sprouts after being watered. This shows that: 1

(a) water makes seeds alive from stone
(b) seeds are non-living
(c) sprouting is random
(d) a living thing can stay inactive until conditions suit it

15. A town has 1,00,000 vehicles each emitting smoke. Which step reduces the problem most? 1

(a) less trees **(b)** more public transport
(c) more roads only **(d)** more cars

16. Which of the following explains day and night on Earth? 1

(a) the Sun moves around the Earth **(b)** Earth rotates on its axis
(c) the Moon blocks the Sun daily **(d)** Earth revolves around the Sun once a day

17. **Assertion (A):** The SI unit of length is the metre. 1
Reason (R): 1 metre is equal to 100 centimetres.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

18. **Assertion (A):** Clear glass is a transparent material. 1
Reason (R): Light passes through transparent materials almost completely.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

19. **Assertion (A):** The sense of touch is not a reliable way to judge temperature. 1
Reason (R): Our skin can be misled by what we touched before, so the same water may feel hot or cold.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

20. **Assertion (A):** Evaporation takes place at all temperatures. 1
Reason (R): Boiling takes place only at a fixed temperature, about 100 degrees C.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. Two beakers contain 100 mL water at 70 °C and 700 mL water at 70 °C. Compare their temperatures and say if the bigger one has more heat. 2

22. Explain why a wet cloth kept in a closed polythene bag does not dry. 2

23. A mixture of 50 g salt and 50 g sand is given. After separation, 49 g of salt and 50 g of sand are recovered. Explain the 1 g loss. 2

24. A student claims that a river is living because it flows and changes shape. Rebut the claim. 2

25. Why does the water rise about one fifth in a jar when a burning candle goes out over water? 2

26. On a map of the solar system, 1 cm shows 150 million km. If the Sun-Earth distance is 150 million km, how far away is Mars at about 228 million km? 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. A mixture of salt, sand, chalk powder, iron filings and wooden chips needs to be separated. Suggest the best sequence. 3

28. A student claims that mushrooms are plants. Give three reasons why they are not. 3

29. Calculate the oxygen present in a room of air volume 50 m^3, assuming 21 per cent oxygen. 3

30. A day on planet X is 10 hours and its year is 5000 hours. How many days are there in its year? 3

31. Explain the role of curiosity, observation and reasoning in the discovery of gravity by Newton. 3

OR

A student suggests classifying animals by colour. Why is this not a good method? 3

32. Why should dairy foods be included in a school child's diet? 3

OR

Explain how a loudspeaker uses a magnet. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. **Case study.** Aarav's tiffin usually has only chips and biscuits. His teacher in Chennai explained that such packaged junk food has a lot of fat, sugar and salt but little else. She asked the class to read food labels and to eat a variety of foods such as dal, vegetables, fruits, rotis and milk. — 5

(i) Which nutrient in dal helps growth?
(ii) Name a nutrient in fruits and vegetables that protects the body from disease.
(iii) What can we learn by reading a food label?
(iv) Why is a mix of foods better than only chips and biscuits?

34. **Case study.** Rohan found an old compass in his grandfather's trunk in Amritsar. The needle always settled pointing in one direction, however he turned the box. When he brought a bar magnet near, the needle moved away from the north. — 5

(i) What does the needle of a compass point towards when undisturbed?
(ii) Name the poles of a magnet.
(iii) Why did the needle move when the magnet came near?
(iv) How can a compass help travellers find directions?

35. Design a model of the solar system for your school science fair describing the materials, scales and observations. — 5

36. Design a survey and experiment to study the effect of music on plant growth. Name the control and variables. — 5

OR

Explain how the loss of habitat leads to loss of biodiversity. Suggest five ways to conserve it. — 5

37. Discuss how traditional Indian meals combine foods to give a balanced diet, using thali as an example. — 5

OR

Describe an activity to show that the strength of a magnet is highest at the poles, and write observations and a conclusion. — 5

38. A student says 'My measurement of 12.4 cm is accurate because I measured only once'. Evaluate and suggest improvements. — 5

OR

Describe an experiment to show that the same material in different forms (flat sheet and a boat shape) behaves differently in water. — 5

END OF QUESTION PAPER — Total: 80 Marks

MASTERMIND